

Christopher Kitras

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Provo, UT, USA

Research Interests

- My research focuses on the resilience of network-enabled embedded systems. I design robust sensor networks that adapt to their environment and repurpose existing technologies to extract actionable information. My systems are intended for deployment on commodity off-the-shelf hardware or pre-existing infrastructure. I am particularly interested in how these fields can be leveraged to enhance the capability of simple systems without adding extra hardware to the architecture.
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Education

- Ph.D. in Electrical and Computer Engineering | April 2027 Brigham Young University, Provo, UT
 - GPA 3.97/4.0
 - B.Sc. in Computer Engineering | December 2021 Brigham Young University, Provo, UT
 - GPA 3.6/4.0
 - BYU Half Tuition Scholarship Spring 2019
 - Dean's List Winter 2021
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Graduate Classes

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| • High Performance Computing | • Cybersecurity & Pen Testing |
| • Wireless Networking | • Cyber-Physical Systems |
| • Exploring SDRs | • Scientific Computing (HPC on Supercomputers) |
| • Robotic Vision | • Computational Creativity |
| • Advanced Digital Systems (FPGA Design) | • Advanced Computer Networks |
| • Privacy in Computing Systems | • Professional Writing |
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Publications

- J. Beard, M. Ball, **C. Kitras**, P. Lundrigan. Long-Term Validation of a Consumer-vs. Industrial-Grade Indoor Radon and Air Quality Monitor at a School in Utah. Sensors. Vol 26, 2026

- E. Sorensen, D. Hales, **C. Kitras**, J. Lofthouse, J. Montierth, P. Lundrigan. If You Build It, Will They Connect? 6 GHz WiFi in a Stadium. IEEE International Conference on Computer Communications and Networks (ICCCN), 2026.
 - **C. Kitras**, J. D. Beard, J. D. Johnston, P. Lundrigan. Sustainable Radon Mitigation through Optimized HVAC Scheduling. IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (CHASE), 2025
 - A. Palacios, D. Harman, **C. Kitras**, E. Kelsey, M. Burnett, W. Harrison, P. Lundrigan. Hidden in Plain Sight: Communicating using Interference. IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN), 2025
 - C. E. Flowerday, P. Lundrigan, **C. Kitras**, T. Nguyen, J. C. Hansen. Utilizing Low-Cost Sensors to Monitor Indoor Air Quality in Mongolian Gers. Sensors. Vol 23, Iss 18, 2023
 - **C. Kitras**, C. Pollan, K. Meyers, P. Lundrigan. Location Verification of Crowd-Sourced Sensors. IEEE International Conference on Computer Communications and Networks (ICCCN), 2023
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Posters/Demos

- **C. Kitras**, C. Pollan, J. Linford, P. Lundrigan. TCAM: Traffic Camera Air quality Monitoring. Air Quality: Science for Solutions (S4S), 2025.
 - **C. Kitras**, C. Pollan, J. D. Beard, J. D. Johnston, P. Lundrigan. Radon Mitigation through Optimized HVAC Scheduling. Air Quality: Science for Solutions (S4S), 2024. **Best Poster Award.**
 - **C. Kitras**, A. Palacios, P. Lundrigan. SSS: Building a Seven Segment Sign. PyCon, 2023.
 - **C. Kitras**, C. Pollan, K. Meyers, P. Lundrigan. Location Monitoring Framework for Citizen Science Sensors. Air Quality: Science for Solutions (S4S), 2023. **Best Poster Award.**
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Experience

- Research Assistant | April 2020—Present | Network Enhanced Technologies Lab at Brigham Young University
 - Localization of Low-Cost Air Quality Sensors
 - Devised a software-based solution for precise device location using traceroute data
 - Streamlined location determination with minimal firmware changes avoiding hardware alterations
 - Created a secure registration process to associate location info with sessions

- Led algorithm creation to monitor routing path changes and detect location shifts
- Low-Cost Air Quality Sensors in Extreme Environments
 - Developed a cost-effective air quality monitor centered on the Particle microprocessor platform to gauge effectiveness of energy-efficient housing versus traditional housing
 - Managed sensor fleets using cloud tools, including executing over-the-air updates
 - Optimized sensor firmware for modular design, allowing for new peripherals without redesign
 - Refined cloud data transfer efficiency by integrating buffering services such as Google PubSub
- Ghost Modulation
 - Audited unconventional protocol for vulnerabilities and implemented security measures by identifying attack vectors where an eavesdropper could sniff and replay data
 - Implemented a lean message authentication code (MAC) that ensures data integrity by hashing the message, a shared secret, and a timestamp
 - Collaborated with a joint team to devise a strategic implementation for an optimized solution
- Radon Mitigation
 - Designed a system to reduce radon levels to healthy limits in a large structure without modifying original architecture
 - Optimized monitor selection, choosing a cost-effective yet high-performance SunRADON device to balance budget and system needs
 - Engineered a Python-based client to interface with an undocumented API lacking official clients
 - Formulated an algorithm to project optimal HVAC scheduling for radon mitigation
- Traffic Camera Air Quality Monitoring Platform
 - Developed a platform that enables ML researchers to access to large datasets to enable real-time model training
 - Compiled a dataset over 3TB full of traffic camera images and the corresponding weather/air quality conditions accessible by a new Python API
 - Designed and maintained web app using MongoDB and Flask instance to store and deliver data

- Integrating a pre-existing ResNet50 model that will create a regression between available PM2.5 values and the traffic camera images
- Latency Shift Keying
 - Successfully communicate with devices across various ranges (from an access point to across the internet) by inducing latency in a device
 - Developed a suite of encoding and decoding strategies that allow an unaffiliated external device to communicate with a device behind a NAT/PAT and firewall
 - Developed three different side channels using this modulation to circumvent traditional router segmentation and security features
 - Created transmitter that repurposes kernel modules to generate traffic pattern to induce slight but observable latency in a router
- Unassociated Device Connectivity to Internet
 - Established downlink from a remote node on the internet to a device that is within close range of a router but not associated
 - Developing MAC for multiple remote hosts to talk with multiple unassociated devices across router
 - Creating a mechanism to predict CPU affinity to provide more reliable service across unwilling target routers

Presentations

- Sustainable Radon Mitigation through Optimized HVAC Scheduling, CHASE 2025, Manhattan, NY, USA
- Location Verification of Crowd-Source Sensors, ICCCN 2023, Honolulu, HI, USA

Teaching Experience

Instructor

- **ECEN 225**: Intro to Computer Systems Laboratory. (Winter 2023)

Teaching Assistant

- **ECEN 426**: Computer Networks. (Fall 2022)
- **ECEN 220**: Fundamentals of Digital Systems. (Fall 2019, Winter 2020)

Professional Activities

- Member of ACM
 - Member of SIGMOBILE
- Member of IEEE

Professional Development

- CyberPOWDER Fellowship, University of Utah, 2026
 - Participated in a semester-long course focused on 5G cellular networks, the POWDER testbed, and wireless systems research ideation; attended an intensive in-person training and experimentation program.
 - Colosseum Young Gladiators Workshop, Northeastern University, 2023
 - Attended in-person program centered on cellular testbeds and an RF raytracing simulator for indoor OTA experiments.
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Open Source Projects

- SSS
 - I helped create a Seven Segment Sign as a fun interactive display for our lab space. It uses over 1100 individual seven segments joined in a 48x24 grid to create a screen that is controlled entirely by our custom software that performs non-interactive demo looping while allowing users to interrupt and change the demo/game at any moment from a web app. This display won **first place** at the BYU IoT Pi Competition.
 - This project was displayed as a demo and a poster at PyCon 2023.